

Quantum Algorithms

Hrishikesh Belagali

January 15, 2026

1 Basics

1.1 Basic Gates

Definition: XOR

Let $a, b \in \{0, 1\}^n$. The **exclusive OR** (XOR) of a and b , denoted by $a \oplus b$, is defined as:

$$(a \oplus b)_i = \begin{cases} 0 & \text{if } a_i = b_i \\ 1 & \text{if } a_i \neq b_i \end{cases}$$

for each $i = 1, 2, \dots, n$.

Definition: FANOUT

Let $a \in \{0, 1\}^n$ and $k \in \mathbb{N}$. The **FANOUT** of a to k copies, denoted by $\text{FANOUT}(a, k)$, is defined as the concatenation of k copies of a :

$$\text{FANOUT}(a, k) = \underbrace{a \| a \| \dots \| a}_{k \text{ times}} \in \{0, 1\}^{nk}$$

The FANOUT is denoted by a solid dot with multiple outputs.

Note: $\text{FANOUT}(a) = \text{CNOT}(a, 0)$ When you have a CNOT with a hollow dot, it means you are copying the negation of the control qubit to the target qubit.

1.2 Universal Collection of Gates

We will first define what it means for a collection of gates to be universal for classical computing.

Definition: Universal gateset

A collection of logic gates is said to be universal for classical computing iff

$$\forall m, n \in \mathbb{N}, \forall f : \{0, 1\}^m \rightarrow \{0, 1\}^n,$$

there exists a Boolean circuit constructed from the gates in the gateset that computes f .

The following gatesets are universal for classical computing:

- {NAND}
- {AND, OR, NOT} (You don't need the OR here)
- {Toffoli}

The proofs can be done by induction on n , the number of output bits.

Definition: Ancilla bits

The **ancilla bits** are extra bits initialized to either 0 or 1 that can be used in the computation of a Boolean function.

1.3 Reversible Gates and Circuits

A reversible gate/circuit is one where you can uniquely recover the input from the output. Naturally, it would make sense for the Boolean function it implements to be reversible.

Definition: Reversible gate/circuit

A logic gate/circuit is said to be reversible if it computes a bijective function from $\{0,1\}^n$ to $\{0,1\}^n$ for some $n \in \mathbb{N}$.

Example

The NOT gate is reversible and its inverse is itself.

Example

The CNOT gate is reversible and its inverse is itself. This makes more sense if you write it as a function:

$$\text{CNOT}(a, b) = (a, a \oplus b)$$

To recover (a, b) from $(a, a \oplus b)$, we can do:

$$(a, (a \oplus b) \oplus a) = (a, b)$$

An irreversible circuit can be made reversible through the use of ancilla bits.

Definition: Register

A register is a collection of bits (or qubits).

Lemma 1.1: Reversible circuit construction using Toffoli gates

For any $m, n \in \mathbb{N}$ and any $f : \{0,1\}^n \rightarrow \{0,1\}^m$, there exists $p \in \mathbb{N}$, a bit string $a \in \{0,1\}^p$ and $g : \{0,1\}^n \rightarrow \{0,1\}^{n+p-m}$ such that: there is a reversible circuit C constructed from Toffoli gates that implements $\{0,1\}^n \times \{0,1\}^p \rightarrow \{0,1\}^m \times \{0,1\}^{n+p-m}$ defined by: $(x, a) \mapsto (f(x), g(x))$. $g(x)$ is called the garbage function.